Reg. No.															

M.Tech.DEGREE EXAMINATION, DECEMBER 2024
OPEN BOOK EXAMINATION
First Semester
21IPC501J - RESEARCH METHODOLOGY
(For the candidates admitted from the academic year2024 - 2025)

Note:
.Specific approval Three text books(Printed or photocopy) recommended for the course.
.Handwritten class notes(certified by the faculty handling the course / Head of the Department).

Time: 3 hours Max. Marks: 100

(5 x 20 = 100 Marks)
Answer **ANY FIVE** Questions

Marks BL CO PO

1. i) The 30 publications and their citations of a researcher is tabulated below. Determine the h-index and g-index of the researcher using plots. Also draw the graphical representation of the data used to deduce the results [5 Marks] 20 3 1 2

Year	Citations
2000	31
2002	47
2003	10
2004	42
2004	35
2005	44
2006	32
2006	9
2008	20
2009	23
2010	17
2011	52
2011	16
2012	41
2012	38
2012	14
2013	24
2014	30
2014	5
2016	27
2016	8
2017	8
2018	41
2018	8
2019	6
2020	5
2021	3
2021	2
2022	1
2022	1

ii) For a Journal 'A' , Calculate the impact Factor for 2023 ,Cite score,SJR, SNIP [5 Marks]

Citations of 2022 = 1050

Citations of 2021 = 1100

No of papers published in 2022=250

No of papers published in 2021=250

iii) Prepare a structured questionnaire for conducting a research on the topic "Understanding the factor influencing the students in selecting Computer science stream Vs Core courses after 12th std [5 Marks]

iv) Identify the type of research for the following [5 Marks]

1. The study of behavioural response of employees to the changes occurring in an organization
2. Number of mortality rate due to various reasons
3. Survey of public response before launching a product
4. Cause for the inflation in a country
5. Customers preference to buy a specific item

2. E. P. Kukula, M. J. Sutton and S. J. Elliott, "The Human–Biometric-Sensor Interaction Evaluation Method: Biometric Performance and Usability Measurements," in *IEEE Transactions on Instrumentation and Measurement*, vol. 59, no. 4, pp. 784-791, April 2010, doi: 10.1109/TIM.2009.2037878 20 2 2 1

i) How will you refer the above article with the following styles as list of references? [5 Marks]

- APA
- IEEE
- Chicago
- AMA
- MLA

ii) How will you cite the above article with the following styles in the content of your paper? [5 Marks]

- APA
- IEEE
- Chicago
- AMA
- MLA

iii) In the context of your chosen field of research, how would you organize an research article? Discuss in detail the significance of each section of the article, and explain how each contributes to overall structure of the article. Use relevant examples.Also suggest a possible national level Journal for communication (Use Domain level references) [10 Marks]

3. i) "Ethics in research is the need of the hour". Justify the statement. [5 Marks] 20 4 3 2

ii) **Scenario :** Kumara Sankar, a student at an university, is eager to publish a research paper to boost his chances of securing admission to a prestigious postgraduate program. Pressurized with time limit and academic requirements, he finds an article published in an international journal on a topic similar to his coursework. Kumara Sankar decides to copy large portions of the article, slightly altering some sentences and substituting a few words, and submits it to a local conference under his name.

The paper gets accepted, and Kumara Sankar is praised by his peers and faculty. However, after few days, one of the participants who is familiar with the original journal article recognizes the copied content and raises the issue with the conference organizers and Kumara Sankar's university.

Read the given scenario and address the following [15 Marks]

1. Immediate action for Kumara Sankar's publication
2. Impact on stake holders
3. Preventive measures to avoid such situations
4. Failure of the author to uphold the ethical responsibility
5. Unethical practices due to pressure on scholars to publish papers
6. Role of plagiarism detection tools and strict policies in deterring such actions

4. i) The following data is related to the measurement of cholestrol content (%) performed in 4 different laboratories w.r.t 4 diet foods. The design adopted is a Randomized Block design. Analyze the data and comment on your result [Given F(0.05) =3.86 for (3,6) D.F] 20 4 4 2

[14 Marks]

Diet	Laboratory			
	A	B	C	D
D1	2	3	4	2
D2	5	3	7	5
D3	4	5	6	4
D4	3	4	5	8

ii) Indicate the need of experimental designs in Education. Distinguish between experimental methods and experimental designs. [6 Marks]

5. i) An official Government population census was conducted and it was reported that in Potheri town ,60% of families have only one child,28% have two children and 12% have three or more children. To verify this, a scholar from SRM decided to conduct his survey in Potheri town and has found that out of 1290 families ,730 had one child and 380 had two children. Determine whether his data supports the results of the Government census. Assume a significance level of 0.05 [14 Marks] 20 4 5 2

ii) Differentiate between research hypotheses and null hypotheses and illustrate your answer with suitable example. [6 Marks]

6. i) Compute the rank correlation coefficient for the following data of the marks obtained by five students in Electrical Engineering and Biology [10 Marks] 20 3 5 2

Electrical Engineering	15	20	28	40	60
Biology	40	30	50	20	10

ii) Draw the block diagram of a traffic monitoring system and explain the role of deep learning algorithm to detect and track vehicles in the traffic monitoring system [10 Marks]

7. "Imagine you're a researcher in the year 2035, where electric vehicles (EVs) have become the primary mode of personal transportation globally. However, this rapid shift has caused a sharp rise in electricity demand, placing immense pressure on power grids worldwide. Your task is to develop a comprehensive research proposal to tackle this issue. Your proposal should: 20 2 2 1

1. Identify a broad research question that frames the challenge.
2. Define three specific research objectives aimed at finding innovative and sustainable solutions.
3. Justify the direction of your research by explaining its importance and expected impact.
4. Highlight at least two disciplines outside of engineering that your research will incorporate and explain how they contribute to addressing the problem.
5. Your approach should consider technological, social, and environmental factors and aim to explore novel solutions beyond those currently discussed in mainstream or academic circles.

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